Lecture 9 - Linearization and Derivative Rules

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Post Exam Recap

1. Solutions to Exam 1 are up on Canvas, I'm happy to answer any questions about it if people want
2. Exams should be graded by tonight or tomorrow night, with curve details to follow after I release the scores
3. One important note so far: follow the instructions in the questions! Many people did not include reasoning for parts b and c of question 2; this will decrease your scores!

Quiz Time!

Let’s get into neat rows and spread out for the quiz!
Just kidding :) no quiz today

Linearization

Best Linear Approximation

Recall that the *derivative* of a function $f(x)$ at a point $x = a$ was equal to the slope of the *best linear approximation* to $f(x)$ at $x = a$.

$$x \approx a \quad \Longrightarrow \quad f(x) \approx f(a) + f'(a)(x - a). \tag{1}$$

Linearization

The linearization of a differentiable function $f(x)$ at a point $x = a$ is the best linear approximation to $f(x)$ at $x = a$ and is equal to

$$\mathscr{L}_{x=a}(x) = f(a) + f'(a)(x - a) \tag{2}$$

Using the Linearization to Approximate I

Using linearizations gives us *the strongest* ways approximate functions in a quick and accurate way.

Example 1: The *logistic population growth* model is a mathematical description of how the size of a population $P(t)$ grows over some time period t with some growth factor k and some carrying capacity N. It assumes that:

$$\frac{dP}{dt} = kP(N - P) \tag{3}$$

Suppose that we have a population $P(t)$, where t is measured in days, which:

1. Starts with $1,000$ individuals So $P(0) = 1000$
2. Has a growth rate of 0.0001 So $k = 0.0001$
3. Has a carrying capacity of $11,000$ individuals So $N = 11000$

What is our best estimate for the population size one year after we started measuring?

Using the Linearization to Approximate II

$$\frac{dP}{dt} = kP(N - P), \quad P(0) = 1000, \quad k = 0.0001, \quad N = 11000, \quad P(1) \approx ? \tag{4}$$

Let's linearly approximate! The linearization of $P(t)$ around $t = 0$ is:

$$\mathscr{L}_{t=0}(t) = P(0) + P'(0)(t - 0) \tag{5}$$

We know $P(0) = 1000$ what about $P'(0)$? We use the $\frac{dP}{dt}$ equation!

$$P'(0) = \frac{dP}{dt}\,|_{t=0} = k \cdot P(0) \cdot (N - P(0)) \tag{6}$$

$$= 0.0001 \cdot 1000 \cdot (11000 - 1000) = \frac{1}{10} \cdot 10000 \tag{7}$$

$$= 1000 \tag{8}$$

And so:

$$P(1) \approx P(0) + P'(0)(1 - 0) = 1000 + 1000 \cdot 1 = 2000 \tag{9}$$

Using the Linearization to Approximate III

Is this an over-approximation ($\mathscr{L}_{t=0}(1) > P(1)$) or an under-approximation ($\mathscr{L}_{t=0}(1) < P(1)$)? What would we need to determine that?
Suppose we know that $P''(0) = 900$. Does that mean our linearization gave an over-approximation or an under-approximation?
Since $P''(0) > 0$, $P(t)$ is *concave-up* at $t = 0$, which means our approximation is an *under* approximation!
(Concave up => graph is above its tangent line)

Using the Linearization to Approximate IV

We can also approximate functions that are:

1. Easy to define but
2. Hard to evaluate

like $f(x) = \sqrt{x}$, around known values.

Example 2: Approximate $\sqrt{25.1}$.

Approximation should always have us think of linearization: what do we need to linearize?

We need:

1. An initial value $x = a$ around which to approximate where
2. We know $f(a)$ AND
3. We know $f'(a)$.

What is a value $a \approx 25.1$ where we know $\sqrt{a}$ and the derivative of the sqrt function at $x = a$?

Using the Linearization to Approximate V

We know that:

$$\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}} \tag{10}$$

so knowing $\frac{d}{dx}\,|_{x=a}\,\sqrt{x}$ is as hard as knowing $\sqrt{a}$. What should a be?

We should set $a = 25$ since $25 \approx 25.1$ AND $\sqrt{25} = 5$ is easy!

Then we get:

$$\sqrt{25.1} \approx \sqrt{25} + \frac{1}{2 \cdot \sqrt{25}}(25.1 - 25) \tag{11}$$

$$= 5 + \frac{1}{2 \cdot 5} \cdot (0.1) \tag{12}$$

$$= 5 + \frac{1}{10} \cdot 0.1 \tag{13}$$

$$= 5 + 0.01 = 5.01 \tag{14}$$

Using the Linearization to Approximate VI

Let's say we also knew that:

$$\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{-1}{4\sqrt{x^3}}. \tag{15}$$

Is our approximation then an over or under approximation?
Well:

$$\frac{d^2}{dx^2}\Big|_{x=5}\sqrt{x} = \left(-\frac{1}{4\sqrt{x^3}}\right)\Big|_{x=5} \tag{16}$$

$$= -\frac{1}{4\sqrt{125}} < 0 \tag{17}$$

So this is an *over* approximation since we are concave down at $x = 5$.

Derivative Rules

Limit Definition is Hard

How can we get that $\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{-1}{4\sqrt{x^3}}$?

$$\frac{d}{dx}\frac{1}{2\sqrt{x}} = \lim_{h\to 0}\frac{\frac{1}{2\sqrt{x+h}} - \frac{1}{2\sqrt{x}}}{h} \tag{18}$$

$$= \frac{1}{2}\lim_{h\to 0}\frac{\frac{\sqrt{x+h}}{x+h} - \frac{\sqrt{x}}{x}}{h} \tag{19}$$

$$= \frac{1}{2}\lim_{h\to 0}\frac{\frac{x\sqrt{x+h}}{x(x+h)} - \frac{(x+h)\sqrt{x}}{x(x+h)}}{h} \tag{20}$$

$$= \frac{1}{2}\lim_{h\to 0}\frac{\sqrt{x^3+hx^2} - \sqrt{(x+h)^2\cdot x}}{hx(x+h)} \tag{21}$$

$$= \ldots \tag{22}$$

GROSS!!

Derivative Rules Rule!

We'll see that a much easier way to compute this derivative involved using *derivative rules* - variants of the sum/product/quotient limit rules we discussed a couple weeks ago.
For a sneak peek:

$$\frac{d}{dx}\frac{1}{2\sqrt{x}} = \frac{d}{dx}\frac{1}{2}\frac{1}{x^{\frac{1}{2}}} \tag{23}$$

$$= \frac{d}{dx}\frac{1}{2}x^{-\frac{1}{2}} \tag{24}$$

$$= \frac{1}{2}\frac{d}{dx}x^{-\frac{1}{2}} \tag{25}$$

$$= \frac{1}{2}\cdot -\frac{1}{2}\cdot x^{-\frac{1}{2}-1} \tag{26}$$

$$= \frac{-1}{4\sqrt{x^3}} \tag{27}$$

Derivative Rules

These will provide a *synthetic* approach to computing derivatives, and are split up similar to our rules for limits: some rules tell us how our operation acts on "simple" things, the rest tell us how our operation acts on "combinations" of things.

Atomic Rules:

- Power Rule: $\frac{d}{dx}x^n$
- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

Sum/Difference Rule

Perhaps the most straightforward derivative rule:

Sum/Difference Rule

The derivative of a sum/difference of functions is the sum/difference (resp.) of the derivatives of the functions:

$$\frac{d}{dx}(f(x) + g(x)) = \left(\frac{d}{dx}f(x)\right) + \left(\frac{d}{dx}g(x)\right). \tag{28}$$

Why? What is the slope of a sum of two linear functions?

$$(f(a) + f'(a)(x-a)) + (g(a)g'(a)(x-a)) = (f(a) + g(a)) + (f'(a) + g'(a))(x-a) \tag{29}$$

Constant Multiple Rule

Similarly:

Constant Multiple Rule

The derivative of a function multiplied by a constant is that constant multiplied by the derivative of the function:

$$\frac{d}{dx}(c \cdot f(x)) = c\frac{d}{dx}f(x) \tag{30}$$

Why? What is the slope of a constant multiple of a linear function?

$$c \cdot (f(a) + f'(a)(x - a)) = (c \cdot f(a)) + (c \cdot f'(a))(x - a) \tag{31}$$

WARNING!

The factor must actually be a constant!

$$\frac{d}{dx}x^2 = 2x \neq x \cdot \frac{d}{dx}x = x \cdot 1 = x \tag{32}$$

Example

We can use this to compute derivatives of functions like:

$$f(x) = 5x^3 + 3x^2 + 2x + 5 \tag{33}$$

Since we already know that

$$\frac{d}{dx}x^3 = 3x^2, \quad \frac{d}{dx}x^2 = 2x, \quad \frac{d}{dx}x = 1, \quad \frac{d}{dx}1 = 0 \tag{34}$$

we get:

$$f'(x) = \frac{d}{dx}(5x^3 + 3x^2 + 2x + 5) \tag{35}$$

$$= \frac{d}{dx}5x^3 + \frac{d}{dx}3x^2 + \frac{d}{dx}2x + \frac{d}{dx}5 \tag{36}$$

$$= \frac{d}{dx}(5 \cdot x^3) + \frac{d}{dx}(3 \cdot x^2) + \frac{d}{dx}(2 \cdot x) + \frac{d}{dx}(5 \cdot 1) \tag{37}$$

$$= 5 \cdot \frac{d}{dx}x^3 + 3 \cdot \frac{d}{dx}x^2 + 2 \cdot \frac{d}{dx}x + 5 \cdot \frac{d}{dx}1 \tag{38}$$

$$= 5 \cdot (3x^2) + 3 \cdot (2x) + 2 \cdot 1 + 5 \cdot 0 \tag{39}$$

$$= 15x^2 + 6x + 2 \tag{40}$$

Power Rule I

What about for more complicated polynomials e.g. what if the degree is > 3?
We know that:

$$\frac{d}{dx}x^3 = 3x^2 \tag{41}$$

$$\frac{d}{dx}x^2 = 2x \tag{42}$$

$$\frac{d}{dx}x^1 = 1 \tag{43}$$

$$\frac{d}{dx}x^0 = 0 \tag{44}$$

Can we see a pattern?
Guess: $\frac{d}{dx}x^n = n \cdot x^{n-1}$.

Power Rule II

Why?

$$\frac{d}{dx}x^n = \lim_{h\to 0}\frac{(x+h)^n - x^n}{h} \tag{45}$$

$$= \lim_{h\to 0}\frac{_x^n + _x^{n-1}\cdot h + _x^{n-2}h^2 + \ldots + _x^0 h^n - x^n}{h} \tag{46}$$

$$= \lim_{h\to 0}(_x^{n-1}\cdot 1 + _x^{n-1}h + \ldots + _x^0 h^{n-1}) \tag{47}$$

$$= _x^{n-1} \tag{48}$$

What is the coefficient on hx^{n-1} in $(x+h)^n$? It's n!

So we know that $\frac{d}{dx}x^n = nx^{n-1}$!

Power Rule III

Power Rule

$$\frac{d}{dx}x^n = nx^{n-1} \tag{49}$$

E.g.:

- $\frac{d}{dx}x^{100} =$
- $\frac{d}{dx}x^{67} =$
- $\frac{d}{dx}x$

Power Rule IV

Note: We technically only showed the limit definition worked for *integer* powers (x^1, x^2, x^3) as opposed to fractional powers, i.e. roots ($x^{\frac{3}{2}}, \sqrt{x}, \sqrt[3]{x^2}$). However, with product rule or inverse rule it's easy enough to show that power rule also works for fractional, and even all real-valued power functions.

E.g. $\frac{d}{dx}x^{\pi} =$

WARNING!!

Power rule only applies to ***POWER FUNCTIONS*** i.e. $f(x) = x^c$ for some constant c.
It does NOT apply to *exponential functions* like $f(x) = c^x$! They may look similar, but their derivatives are VASTLY different!

Exponential Rule I

What is the derivative of an exponential function?

$$\frac{d}{dx}c^x = \lim_{h\to 0}\frac{c^{x+h}-c^x}{h} \tag{50}$$

$$= \lim_{h\to 0} c^x \frac{c^h-1}{h} \tag{51}$$

$$= \left(\lim_{h\to 0}\frac{c^h-1}{h}\right)c^x \tag{52}$$

it is just some constant multiple of the original function!

What is the constant factor $\lim_{h\to 0}\frac{c^h-1}{h}$?

Exponential Rule II

Define a constant e so that $\lim_{h\to 0} \frac{e^h - 1}{h} = 1$ (i.e. $\frac{d}{dx} e^x = e^x$). Then:

$$\lim_{h\to 0} \frac{c^h - 1}{h} = \lim_{h\to 0} \frac{(e^{\ln(c)})^h - 1}{h} \tag{53}$$

$$= \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{h} \tag{54}$$

$$= \lim_{h\to 0} \frac{\ln(c)}{\ln(c)} \frac{e^{\ln(c)h} - 1}{h} \tag{55}$$

$$= \ln(c) \lim_{h\to 0} \frac{e^{\ln(c)h} - 1}{\ln(c)h} \tag{56}$$

$$= \ln(c) \lim_{z\to 0} \frac{e^z - 1}{z} = \ln(c) \cdot 1 = \ln(c) \tag{57}$$

Exponential Rule III

This gives us:

Exponential Rule

The derivative of an exponential function, i.e. c^x for a constant c is:

$$\frac{d}{dx}c^x = \ln(c)c^x \tag{58}$$

Exponential functions growing rapidly => it's derivative grows very large very quickly.
Very different behavior to that for power rule $\frac{d}{dx}x^c = cx^{c-1}$.

Exponential Rule IV

What is the n'th derivative of an exponential function c^x?
First derivative: $\ln(c)c^x$.
Second derivative: $\frac{d}{dx}(\ln(c)c^x) = \ln(c)\frac{d}{dx}c^x = \ln(c)^2c^x$
Third derivative: $\ln(c)^3c^x$.
...
n'th derivative: $\ln(c)^nc^x$.

Worksheet Time!

Let’s get into groups and work on the worksheet!

End of Class

That’s all folks!